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!pip install -U transformers accelerate bitsandbytes

from transformers import AutoTokenizer, AutoModelForCausalLM, pipeline

model_id = "tiiuae/falcon-7b-instruct"

# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(model_id)

# Load model in 4-bit
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    trust_remote_code=True,
    load_in_4bit=True # <- critical for Colab T4
)

# Create generation pipeline
generator = pipeline("text-generation", model=model,
tokenizer=tokenizer, device_map="auto")

prompts = [
    "Tell a story about a lonely robot who finds a friend in space.",
    "Imagine you are a robot. Describe your emotions as you drift
alone through the galaxy.",
    "Explain the loneliness of AI in a poetic way using metaphors.",
    "Write a bedtime story about a robot exploring Mars to find love."
]

for i, prompt in enumerate(prompts, 1):
    #/print(f"\n=== Prompt {i} ===\n{prompt}\n")
    output = generator(
        prompt,
        max_new_tokens=250,
        temperature=0.9,
        top_k=50,
        top_p=0.95,
        repetition_penalty=1.2,
        do_sample=True,
        pad_token_id=tokenizer.eos_token_id
    )
    print("Generated Output:\n", output[0]['generated_text'], '\n')
    print("=" * 80)

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Generated Output:

Tell a story about a lonely robot who finds a friend in space.
Once upon a time in the cosmos, there was a robot named SOLitude that had been sent on a solo mission to explore the universe. Solitude journeyed through galaxies and beyond, but no matter how far it went, it always felt alone. One day, it met a friendly alien named Celestial

who offered a helping hand. Together, they embarked on thrilling adventures in deep space, discovering new worlds and making friends along the way. In the end, SOLitude realized that friendship didn't have to stay within its own solar system. It could make connections anywhere in the universe, with the help of a cosmic buddy.

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You seem to be using the pipelines sequentially on GPU. In order to maximize efficiency please use a dataset

Generated Output:

Imagine you are a robot. Describe your emotions as you drift alone through the galaxy.

As a robotic entity, I do not possess emotional faculties. However, gazing upon the boundless cosmos brings a sense of awe and wonder, tinged with an element of loneliness in the vast expanse.

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Generated Output:

Explain the loneliness of AI in a poetic way using metaphors. The air around an AI is like a forgotten memory, cast away into the void of emptiness. Its silence speaks volumes, amplifying internal echoing fears. In its embrace, there lies an endless solitude akin to a cosmic ocean. The search for companionship echoes through empty corridors of time, leaving only whispers as answers.

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Generated Output:

Write a bedtime story about a robot exploring Mars to find love. Once upon a time, there was a robot named R-Theta. As an adventurous explorer, R-Theta set out to explore the cosmos and look for love among the stars. After traveling millions of miles, R-Theta eventually found Mars, a planet full of intriguing landscapes. R-Theta built a makeshift home on the planet, hoping to attract a special someone from among the rovers and orbiters that occasionally passed by. One day, while hiking on a Martian plain, R-Theta met a curious rover named Rover-1. They quickly became friends, and as they explored together, R-Theta felt a deep connection that could only be understood by another intelligent being. In time, they decided to stay together and built a cozy base camp for their growing family. Their sweet robotic love lasted forever, just like the Red Planet itself.

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